MILITARY SPECIFICATION

POWER PACKAGE, EMERGENCY AC/DC GENERATOR, RAM AIR TURBINE DRIVEN

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an emergency power package consisting of a ram air turbine, brushless AC/DC generator, voltage regulator, and a hydraulic pump to be used on the A-7D/E and TA-7C aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys

FSC 6135

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SPECIFICATIONS CONT

Military

MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-P-7858	Pump, Hydraulic, Power Driven, Fixed Displacement
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-8775	Hydraulic System Components, Aircraft and Missiles, General Specification for
MIL-F-8815	Filter and Filter Elements, Fluid Pressure, Hydraulic Line, 15 Micron Absolute and 5 Micron Absolute Type II Systems, General Specification for
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-P-19692	Pumps, Hydraulic, Variable Delivery, General Specification for
MIL-C-22751	Coating System, Epoxy-Polyamide, Chemical and Solvent Resistant, Process for Application of
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

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SPECIFICATIONS CONT

Military

MIL-G-27617	Grease, Aircraft, Fuel and Oil Resistant
MIL-S-45909	Stud, Locked In, Ring Locked Serrated
MIL-I-45910	Insert, Screw Thread Locked in and Ring Lock, Serrated
MIL-I-45932	Insert, Screw Thread, Thin Wall, Locked In, General Specification for
MIL-E-81910	Electric Power and Control Equipment, Aircraft, Testing, General Specification for

Naval Air Systems Command

AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification and Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	General Requirements for Electronic Equipment

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SPECIFICATIONS

Military

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MS 33540	Safty Wiring and Cotter Pinning, General Practices for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Precedence. The contract shall have precedence over any specification. Whenever the requirements of the general specification and the detail specification or standard conflict, the requirements of the detail specification or standard shall govern. Any deviation from this specification or from subsidiary specifications or standards, where applicable, are subject to approval in writing of the Qualifying Activity.

3. REQUIREMENTS

3.1 Detail Specifications. The individual item requirements shall be as specified herein.

3.2 Qualification. Ram air turbine power packages furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for opening of bids (see 4.3 and 6.4).

3.3 Selection of specifications, standards, and materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification shall be selected in accordance with MIL-STD-143, except as provided in paragraph 3.4.1.

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3.3.1 Standard parts. Military Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their military part number. Any standard part, of the same part number, shall be interchangeable as applied within the power package. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by standard AN or MS parts without alteration and provided the corresponding standard part numbers are referenced in the parts list and on the contractor's drawings. In the event there is no suitable standard part available as of the date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3.2 Materials. Materials used in the manufacture of the power package shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the qualifying Government activity and contain provisions for adequate tests. The use of the contractor's specification does not constitute waiver of Government inspection.

3.3.2.1 Metals. Metals shall be of a corrosion-resistant type or suitably treated to resist corrosion.

3.3.2.1.1 Dissimilar metals. Dissimilar metals, as defined by MIL-STD-889 shall not be used in contact with each other. Where such contact is unavoidable, the metals shall be protected against electrolytic corrosion. When protection is used, it shall be of such a type that a low impedance path is offered to radio frequency currents.

3.3.2.1.2 Magnesium alloy parts. The use of magnesium is prohibited unless specifically approved for each application by the using service. Where approved, magnesium alloy parts shall be surface treated for protection against corrosion in accordance with MIL-M-3171. Use of magnesium is not permitted in hydraulic components.

3.3.2.1.3 Castings. Castings shall be in accordance with MIL-C-6021 class 1A, grade B.

3.3.2.2 Nonmetals. Nonmetals when used, including plastics, fabrics, and protective finishes, shall be moisture resistant,

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and shall not support fungus growth, or shall be treated with a fungicidal agent acceptable to the procuring agency. The nonmetals shall not support combustion, and shall not be adversely affected by weather, aircraft fluids, temperatures, and ambient conditions encountered during operation of the aircraft. Nonmetals may be treated to conform to this requirement.

3.3.2.3 Finishes and protective coatings.

3.3.2.3.1 Aluminum alloy parts. Aluminum alloy parts shall be anodized in accordance with MIL-A-8625. The aluminum oxide film deposited by the treatment shall be removed from actual contact area of all surfaces required to act as a path for electric current and from local areas under screws, nuts, or other hardware used for assembly or mounting purposes, to provide an adequate bonding connection. Aluminum alloys which do not anodize satisfactorily shall be treated with a chemical coating according to MIL-C-5541. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.3.2.3.2 Steel parts. All exterior steel parts and other steel parts subject to corrosion and not in constant contact with oil shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable. In cases where cadmium plating is considered impractical or undesirable, an equivalent treatment which is acceptable to the procuring agency shall be provided.

3.3.2.3.3 Protective coating. An epoxy-polyamide coating in accordance with MIL-C-22751 shall be applied to the exterior surfaces of the emergency power package.

3.3.3 Color. The color of the emergency power package shall be white, color No. 17875, according to FED-STD-595.

3.3.4 Lubricants. Lubricants shall conform to MIL-G-27617, MIL-STD-838 or MIL-L-23699. Silicon base grease shall not be used.

3.3.5 Hydraulic oil. Hydraulic oil used in the pump shall conform to MIL-H-5606.

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3.3.6 Encapsulation. Encapsulation and potting shall be in accordance with MIL-STD-454, Requirement 47.

3.4 Design and construction. The emergency ram-air driven hydraulic-electric power package shall provide means for utilizing ram-air to generate emergency hydraulic pressure and AC/DC electrical power when the package is extended into the airstream. The design shall be consistent with good aircraft accessory practice.

3.4.1 Reliability. The contractor shall conduct a reliability program using MIL-STD-785 as a guide. The ram air turbine power package shall have a reliability of at least 100 hours Mean-Time-Between-Failures (MTBF) while operating in an environment at least as severe as that specified herein. This requirement shall be attainable at any time during the required operating life and after exposure to the specification storage conditions. Further, the power package reliability shall not be degraded below that required in MIL-STD-785 as a result of normal field usage and maintenance. Reliability analysis shall be submitted in confirmation of the MTBF design objective. If this analysis is performed under the procurement, it shall be updated after each of the following applicable design and development milestones:

1. Technical proposal
2. Completion of final design review
3. After each major change from the original design

3.4.2 Simplicity. Simplicity of design resulting from the use of the same part for as many applications as possible in a given model or various models is highly desirable.

3.4.3 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.4.4 Maintainability. The design shall provide for ease of inspection, testing, disassembly, maintenance, repair and reassembly, preferably without the need for special tools or fixtures as specified in AR-10. Component parts shall be

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as foolproof as possible to avoid incorrect assembly which would result in damage or malfunction, or involve safety of flight. The package design shall permit easy replacement on the airplane of a generator/voltage regulator, pump, or turbine.

3.4.5 Service life. The emergency power package shall not require major servicing (removal from the airplane) in less than 250 hours of normal aircraft operation. The overhaul period of the package shall not be less than 100 hours of package operation. A time totalizing meter that will measure 10,000 hours shall be provided in accordance with MIL-M-7793.

3.4.6 Workmanship. All machined surfaces shall have a smooth finish and all details of manufacture including the preparation of parts and accessories, shall be in accordance with good practice for high quality aircraft equipment. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts, plating, coatings, painting, riveting, clearance between soldered connections, ruggedness, and freedom of parts from burrs and sharp edges.

3.4.7 Lubrication. The complete package shall be lubricated during assembly in accordance with MIL-STD-838 and shall not require additional lubrication during it's operating life. The hydraulic pump is to be filled with fluid conforming to MIL-H-5606 prior to shipment from the vendor facility. When shipped the unit must meet the requirements of five years storage in accordance with 3.5.

3.4.8 Cooling. Ram air shall provide cooling air to the generator to maintain temperatures below the maximum safe operating value when subjected to the temperature altitude requirements of 3.11.2.

3.4.9 Dimensions. The ram air turbine power package shall conform to the dimensions as shown in the applicable drawing.

3.4.10 Weight. The maximum weight of the complete power package shall not exceed 42 pounds.

3.4.11 Design changes. After approval of the design proposal, it shall be the responsibility of the manufacturer to coordinate with the purchasing activity if he is at any time

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in doubt as to compliance with or clarity of applicable specifications. After completion of qualification tests, the manufacturer shall make no change in any part without approval of the qualifying agency.

3.5 Storage. The ram air turbine power package shall perform satisfactorily after five years storage in world wide climate conditions as specified in MIL-STD-210.

3.6 Screw threads. Screw threads shall conform to the requirements of MIL-S-7742 or MIL-S-8879. All threads in non-ferrous materials shall employ threaded inserts. Load carrying inserts shall be in accordance with MIL-I-45932 or MIL-I-45910. Studs, if used, shall conform to the requirements of MIL-S-45909. All internal and external threaded parts shall be positively locked by self locking nuts, safety wiring, staking or other approved methods. Staking shall be used only when parts are permanently assembled.

3.7 Safety wiring and staking. Accidental loosening of screws and screw parts and other connections shall be prevented by safety wiring, staking, or other approved methods in accordance with MS33540. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.8 Electrical connections. Terminations shall be as shown on the detail drawing. Connectors and leads shall be secured in a reliable manner to prevent contact with moving parts or chafing by contact with stationary parts. All leads shall be permanently marked with the designating letters.

3.9 Identification of product. Each component of the ram air turbine power package shall be marked with the manufacturer's part number and the manufacturer's name or trademark in accordance with MIL-STD-130.

3.9.1 Nameplate data. Nameplates conforming to MIL-N-18307 shall identify the major components and shall be securely attached to a fixed accessable portion of each unit.

3.10 Performance. The ram air turbine power package will provide aircraft power in a hydraulic and/or electrical emergency situation. The requirements shall be maintained throughout 100 hours of package operation as recorded on the time totalizing meter.

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3.10.1 Emergency power package. The ram air turbine power package shall start and operate, as required, under the environmental conditions of paragraph 3.11 and any combination thereof. It shall also operate, as required when exposed to inflow angle variations of $\pm$ 18° caused by varying aircraft angle of attack.

3.10.1.1 Speed regulation. The rotational speed of the turbine shall correspond to a generator frequency of 400 $\pm$ 40 Hertz. The speed regulating system shall be compatible with the pressure regulator gain characteristic as shown in Figure 5.

3.10.1.2 Normal rated output. The package shall supply, as a minimum, rated hydraulic and electric outputs to the aircraft systems as required by paragraphs 3.10.1.5 and 3.10.3. This power shall be available over the ranges of airspeeds and ram temperatures of Figures 2 and 3 at the altitudes specified. Steady state power output shall be provided as follows:

(a) At airspeeds of 116 knots and above - 1100 watts electrical power, AC and/or DC, as defined in paragraph 3.10.3 plus the capability of sustaining the system and flow sensitive pressure regulator load at sea level and at pump inlet temperature and pressures as defined in paragraph 3.10.2. Power at various altitudes and temperatures shall be furnished as required by Figure 2.

(b) At airspeeds of 126 knots and above - 1250 watts electrical power as defined by paragraph 3.10.1.2.a plus hydraulic system pressure versus airspeed per Figure 4 at sea level and at the airspeeds, altitudes, and temperatures shown in Figure 2.

(c) At airspeeds of 187 knots and above - Hydraulic output as required by paragraph 3.10.1.5 plus continuous rated electrical output as defined in paragraphs 3.10.3.1 and 3.10.3.2 at sea level and at the airspeeds and altitudes specified in Figure 2. This turbine power requirement is necessary at all airspeeds above the minimums specified in Figure 2, to and including the maximum airspeeds specified thereon.

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3.10.1.3 Overload outputs. The ram air turbine shall deliver sufficient power so that hydraulic and electric overloads are available as specified over the ranges of airspeeds in Figure 2. The ram air turbine shall provide, as a minimum, sufficient power at airspeeds of 214 knots and above at all altitudes specified in Figure 2 to maintain:

(a) Simultaneous AC overload as defined in paragraph 3.10.3.1.1.2 and hydraulic load as defined in paragraph 3.10.1.5 plus normal rated DC load. The AC voltage transients shall be within those limits specified in Figure 9, and the DC voltage transients shall be within the limits of Figure 6.

(b) Any one phase of the AC generator short-circuited and with rated DC load plus hydraulic pump load as defined in paragraph 3.10.1.5. The DC voltage shall not be less than 12 volts.

(c) Simultaneous DC overload as defined in paragraph 3.10.3.2.1.2 and any hydraulic load due to the pump and pressure regulator characteristics required by paragraph 3.10.1.5 plus rated AC load. AC and DC voltage limits shall be as specified in 3.10.1.3.a.

(d) DC power source short-circuited with at least 90 amperes DC current for 5 seconds plus rated AC generator load and hydraulic pump load per paragraph 3.10.1.5. The AC voltage shall not be less than 90 volts.

3.10.1.4 Start and acceleration. The ram air turbine shall be capable of starting and accelerating the hydraulic pump and AC/DC generator to the speeds required by the loads defined in the following paragraphs. Start shall be accomplished with ambient temperatures of -55°C to +85°C and with hydraulic system temperatures of -40°C, to +135°C. The initial rotation (breakout of the turbine) will be at least 15 knots less than the values below and subsequent to breakout, a continuous acceleration will occur.

(a) At airspeeds of 116 knots and above - suddenly applied with loads as defined in paragraph 3.10.1.2.a.

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(b) At airspeeds of 126 knots and above - suddenly applied with loads as defined in paragraph 3.10.1.2.b.

(c) At airspeeds of 214 knots and above - suddenly applied with hydraulic load as specified by paragraph 3.10.1.5 plus AC overload of 200 percent at a power factor of 0.8 lagging and DC loads as required by paragraph 3.10.3.2.1.

(d) At airspeed of 187 knots and above - suddenly applied with loads as defined in paragraph 3.10.1.2.c.

(e) At airspeeds of 131 knots and above - suddenly applied with loads described in paragraph 3.10.1.4.b the turbine shall start and accelerate to rated rpm within 8.5 seconds. Hydraulic system temperatures will be at +5°C to +120°C except for the emergency power package and reservoir areas shown in Figure 1 which will be at -55°C.

3.10.1.5 Hydraulic power requirements. The hydraulic power requirements as seen by the turbine depend upon:

(a) The total system per Figure 1 and it's losses.

(b) The operational range of the flow sensitive relief valve as per Figure 5.

(c) The minimum allowable system actual gage pressure requirements as per Figure 4.

(d) The manufacturing tolerances, temperature tolerances, wear tolerances and overall efficiency of the pump.

3.10.2 Emergency hydraulic system. The hydraulic system is shown schematically in Figure 1, which depicts tube bends and identifies system components. The emergency hydraulic system ties into the primary control hydraulic system which consists of a pump, filters for the pump pressure and case drain flows, a number of servo valves and actuators, and the necessary connecting lines. The system is a 3000 psig class, Type II. Return pressure is described in paragraph 3.10.2.3. The normal operating temperature range of the fluid in the pump suction and pressure lines will be +5°C to +120°C. Components must function between -55°C and +135°C.

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3.10.2.1 Flow sensitive pressure regulator. The flow sensitive regulator has the characteristics specified by Figure 5, and the purpose of having this valve installed in the system is to:

(a) Bypass to return any flow in excess of that required by the system while maintaining the pressures required by Figure 4.

(b) Prevent stalling of the pump when no hydraulic flow is required by the system.

(c) Reduce, to a minimum, the hydraulic load on the turbine during acceleration.

3.10.2.2 Hydraulic pump. The variable delivery pump shall be classified as a Type II hydraulic component in accordance with MIL-H-8775. Performance specifications are as follows.

Rated delivery	7.6 gpm
Rated speed	12,650 rpm
Rated pressure	3,000 psi
Fluid	MIL-H-5606
Rotation	Clockwise-viewing pump from shaft end

Performance characteristics required are specified by Figure 10. The pump shall furnish to the airplane system as a minimum the pressure required by Figure 4 and the flow required by Figure 5 when operating in conjunction with the flow sensitive pressure regulator of paragraph 3.10.2.1. Pump outlet pressures in every situation are higher than those required by the system due to characteristic pressure drops across the valve and hydraulic lines. Design shall be in accordance with the following subparagraphs, and the pump shall operate as an integral part of the hydraulic system defined in paragraph 3.10.2. The pump shall be capable of providing pressures to 3150 psig, and withstanding a case proof pressure of 500 psig. For the design of the hydraulic pump, all pressures herein are gage. Model number AP054-43, part number 53010, manufactured by the Abex Coporation is currently used.

3.10.2.3 Inlet pressure. The hydraulic return pressure will be maintained at 85 to 100 psig at the manifold (Item 29 in Figure I). Inlet pressure is equal to return pressure less line pressure drop. The pump shall be designed to accept

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the cavitation associated with these inlet pressures to -55°C as well as the cavitation which occurs during high flight speed (short spin-up time at start-up).

3.10.2.4 Pump capacity. Due to the flow-pressure characteristics of the pressure regulator, fluid flow is a primary consideration in designing the pump. As flow varies from a minimum flow pump to a maximum flow pump, there will be a flow variation resulting from all manufacturing and environmental tolerances. In addition to the production tolerances, flow decay as a result of use shall not provide less than the pressure as specified in Figure 4 at airspeeds from 115 to 630 knots, while not exceeding the maximum AC generator frequency. Rated capacity is 7.6 gpm, 3000 psi, and 12,650 rpm.

3.10.2.5 Pump and tubing leakage. There shall be no external leakage from the pump and hydraulic tubing within the package when pressures from zero to 200 psig are applied at the inlet part or when proof presssure is applied at the outlet part. Leakage at the shaft seal drain part shall not exceed 5 ml/hr during operating or static conditions of the pump.

3.10.2.6 Contamination. The pump shall operate throughout it's service life in a hydraulic system, which uses a 15 micron absolute filter in accordance with MIL-F-8815.

3.10.2.7 Capstrip seals. If capstrip seals or similar devices are used in the hydraulic pump they shall incorporate grooves as the means of preventing hydraulic fluid blow-by throughout the pressure operating range.

3.10.3 AC/DC generator and regulator.

3.10.3.1 AC generator and regulator. The ram air turbine power package shall incorporate a brushless AC generator, and a voltage regulator with an integral underfrequency protector.

3.10.3.1.1 Rating. The AC generator shall be a self-excited, 3 phase, 4 wire, wye connected, 115/200 volt, self-cooled, brushless unit with a minimum continuous rating of 2.5KVA at 0.8 lagging to unity power factor, over a frequency range of 400 ± 40 Hertz. The frequency to voltage (F/V) relationship over the frequency range is shown in Figure 7.

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3.10.3.1.1.1 Continuous capcity. The generator shall be capable of continuously delivering it's rated capacity over the specified airspeed range and at power factors of 0.8 lagging to 1.0.

3.10.3.1.1.2 Overload capacity. The generator shall be capable of delivering 150 percent rated load at 0.8 lagging to unity power factor for 2 minutes and 200 percent rated load at 0.8 lagging to unity power factor for 5 seconds while operating within 400 + 40 Hertz. AC and DC overloads will be applied separately and not concurrently.

3.10.3.1.1.3 Short circuit capacity. The generator shall be capable of supplying a minimum of 300 percent rated load current for five seconds without damage during a single phase line-to-neutral or line-to-line short circuit. The minimum DC voltage during an AC short circuit shall be 12 volts.

3.10.3.1.2 Voltage limits. The steady state voltage shall be 115 + 7 volts. Below 350 Hertz, the underfrequency protector shall limit the voltage to approximately 40 volts or below. Transient voltages that result from the application or removal of loads shall conform to the transient voltage envelope of Figure 9, and not exceed 115 $\pm$ 7 volts for longer than 0.2 seconds.

3.10.3.1.3 Voltage modulation. The voltage modulation shall not exceed an amplitude of 3.5 volts, as measured peak-to-valley, difference between the minimum voltage reached and the maximum voltage reached during any one second. The frequency characteristics of the voltage modulation shall conform to the envelope limits shown in Figure 11.

3.10.3.1.4 Phase displacement and voltage unbalance. A 25 percent load unbalance shall not cause more than a 6 percent voltage unbalance. A 20 percent load unbalance shall not cause more than a 4 percent voltage unbalance. The displacement between adjacent phases of the fundamental voltage shall be 120 $\pm$ 4°.

3.10.3.1.5 Wave form. The crest factor for each phase voltage wave form shall be 1.41 + 15 percent. No single harmonic shall exceed 5 percent of the fundamental. The total harmonic content shall not exceed 8 percent.

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3.10.3.1.6 Frequency limits. The generated frequency shall not exceed 400 ± 40 Hertz over the airspeed range as specified herein and from no load to rated load for all steady state conditions. The frequency shall not exceed 400 ± 40 Hertz for longer than 2 seconds when load is supplied or removed over the range of airspeeds specified for rated loads.

3.10.3.1.7 Frequency modulation. Variations of frequency amplitude due to frequency modulation shall be within a band of ± 2 Hertz during any one minute period. The ± 2 Hertz band shall be within 400 ± 40 Hertz for all steady state operating conditions specified herein. The rate of frequency change due to frequency modulation shall not be greater than 25 Hertz per second.

3.10.3.2 DC generator. The ram air turbine power package shall incorporate a brushless DC generator.

3.10.3.2.1 Rating. The DC generator shall be a 28 volt, self-cooled, brushless unit with a minimum continuous rating of 30 amperes.

3.10.3.2.1.1 Continuous capacity. The generator shall be capable of continuously delivering it's rated capacity over the specified airspeed range.

3.10.3.2.1.2 Overload capacity. The generator shall be capable of delivering 150 percent rated load for 2 minutes and 200 percent rated load for 5 seconds. AC and DC overloads shall be applied separately and not concurrently.

3.10.3.2.1.3 Short circuit capacity. The generator shall be capable of supplying short circuit current as specified in Figure 8. The minimum AC voltage during a DC short circuit shall be 90 volts.

3.10.3.2.2 Voltage limits. The steady state voltage shall be 24 to 30 volts. Transient voltages that result from the application or removal of loads shall conform to the transient voltage envelope of Figure 6, and not exceed the limits of 24 to 30 volts for longer than 0.5 seconds.

3.10.3.2.3 Ripple voltage. The AC peak of ripple voltage to average DC voltage shall not exceed 1.5 volts for either polarity.

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3.10.3.3 Dielectric strength. For circuits of 50 volts and less, the power package shall be capable of withstanding a potential between each terminal and case of 500 volts rms for 1 minute or 600 volts rms for 1 second. For circuits of over 50 volts, the power package shall be capable of withstanding twice the rated voltage plus 1,000 volts rms for 1 minute, or 120 percent of the 1 minute voltage for 1 second. If capacitors are used, they shall be capable of withstanding a test voltage of twice the maximum peak voltage encountered during normal operating conditions or a minimum of 100 volts, whichever is greater.

3.10.4 Ram air turbine. The ram air turbine shall be capable of utilizing ram air at the installed airspeeds of Figure 2 and the environmental conditions of paragraph 3.11 to supply adequate shaft horsepower to the generator set and hydraulic pump to satisfy all of the requirements of paragraph 3.10.1 and all subparagraphs thereunder.

3.10.4.1 Ram air turbine governor. The rotational speed control of the turbine shall be furnished as an integral part of the package, and shall govern as required by paragraphs 3.10.4.1.1 and 3.10.4.1.2.

3.10.4.1.1 Steady state governoring. The maximum governed speed shall be 440 Hertz which corresponds to the AC generator frequency under any load condition imposed by paragraph 3.10.1 and it's subparagraphs. When the available turbine power is less than the rated hydraulic load plus the applied electrical load, the turbine shall slow down to an equilibrium speed where the available turbine power equals the power demand as determined by the hydraulic system and the applied electrical load. Under this condition the frequency shall not be less than 360 Hertz. When the available turbine power is in excess of any requirement set for the herein, the turbine speed shall tend toward a speed such that the system hydraulic pressure shall be as close to 3000 psig as possible within the limits of Figure 5.

3.10.4.1.2 Transient governing. Transient speeds shall not vary the frequency by more than 10 Hertz under the following conditions:

(a) A change in airspeed from 270 knots to 592 knots in 40 seconds.

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(b) After a step change in turbine load from the minimum hydraulic load of paragraph 3.10.1.2.a and zero electrical load to the conditions in paragraph 3.10.1.2.c at a turbine inlet ram pressure 5 percent higher than required for steady state conditions.

(c) After the application of any overload condition specified in paragraph 3.10.1.3.

(d) Steady state performance shall be attained within 1.5 seconds after any of the changes specified above.

3.11 Environmental requirements. The ram air turbine power package shall operate in accordance with the requirements of this specification when subjected to the following environmental conditions or natural combination of conditions specified herein. Performance requirements shall be met before and after environmental tests.

3.11.1 Temperature shock. As specified in 4.8.7.3.

3.11.2 Electromagnetic interference. As specified in MIL-E-81910.

3.11.3 Sand and dust. As specified in MIL-E-81910.

3.11.4 Shock. As specified in MIL-E-81910.

3.11.5 Vibration. As specified in MIL-E-81910 except that operation during test is not required.

3.11.6 Salt fog. As specified in MIL-E-81910.

3.11.7 Fungus. As specified in MIL-E-81910.

3.11.8 Humidity. As specified in MIL-E-81910.

3.11.8.1 Moisture. Rain, dew, fog and frost, and contaminated water incidental to aircraft washdown with the power package retracted.

3.11.9 Explosive atmosphere. The unit shall not cause an explosion when operated in an explosive atmosphere.

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3.11.10 Operating position. The unit shall operate in any position.

4. QUALTIY ASSURANCE PROVISIONS

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. Records shall describe all failures and correction actions occurring during tests and all discrepancies found during inspections.

4.2 Classification of tests. The inspection and testing of ram air turbine power packages shall be classified as follows:

4.2.1 Qualification tests. Qualification tests are those tests accomplished on ram air turbine power packages submitted for qualification as a satisfactory product.

4.2.2 Acceptance tests. Acceptance tests are those tests accomplished on ram air turbine power packages manufactured and submitted for acceptance under a production contract.

4.2.3 Qualification verification tests. Qualification verification tests are those tests conducted on production samples to verify that the design is identical to that qualified and that production units meet all the requirements of the specification.

4.3 Qualification tests. A ram air turbine power package will be considered qualified and placed on the Qualified Products List (QPL) if qualification test samples successfully complete the qualification tests designated by Table I essentially in the order listed and upon acceptance by the qualifying activity. The ram air turbine power packages will be retained on the Qualified Products List as long as qualification verification samples delivered as required below (paragraph 4.5) continue to demonstrate conformance to the specification.

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4.3.1 Qualification test sample. The qualification test sample shall consist of:

(a) Two (2) ram air turbine power packages as described by this specification

(b) A mating plug for every connector used

(c) Two (2) sets of wiring diagrams, detail assembly drawings, outline drawings, parts list, and operating instructions.

(d) Two (2) copies of the material to be included in the installation instruction to be packaged with each production unit as an example of how this material is to be printed, protected, and attached to the unit.

4.4 Acceptance tests. The contractor shall be responsible for accomplishing the inspection tests specified herein. When inspection is conducted at the contractor's plant, all inspection and testing shall be subject to the approval of the Government Inspector. The contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Government Inspector. The contractor shall maintain a record available to the Inspector, showing the results of all inspection tests and signed by an authorized representative of the contractor or laboratory. Acceptance or approval during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.5 Qualification verification tests. When qualification verification is required by the production contract, the sample(s) will be selected at random by the government inspector and forwarded to the qualifying activity. To meet delivery requirements for the first fifty (50) ram air turbine power packages, fifty-one (51) must successfully meet the acceptance tests of Table I. Fifty (50) units will be shipped as directed by the contract and one (1) randomly selected sample will be forwarded to the qualifying activity. To meet the delivery for each succeeding one hundred and fifty (150) ram air turbine power packages, one hundred and fifty-one (151) must successfully meet the required acceptance tests. One hundred and fifty (150) will be shipped as directed by the contract, and one (1) unit selected and forwarded as above. Once qualified, production power packages successfully passing the acceptance tests are to be accepted until qualification is officially

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rescinded. Delivery of production ram air turbine power packages is not to be delayed pending results of tests conducted on qualification verification samples. All qualification verification testing is to be conducted by the qualifying activity. With each qualification verification sample, the Government Inspector shall furnish the contract number and the serial number of all ram air turbine power packages in the production lot from which the verification sample was selected.

4.6 Retest of qualification samples. Ram air turbine power pacakges which have been rejected or returned to the manufacturer for any reason during qualification tests may be reworked or have parts replaced to correct defects. Before submitting them, full corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. Where qualification tests are conducted under the auspices of the manufacturer, the qualifying activity shall be advised upon failure of the qualification sample and of the action taken by the manufacturer with regard to failure.

4.6.1 Retest of qualification verification samples. Ram air turbine power packages may be subjected to any or all requirements and tests of this specification to assure continued compliance with the qualification requirements. Those which are found to be defective for any reason shall be returned to the manufacturer for rework or parts replacement. Full particulars concerning the manufacturer's analysis of the defect and the corrective action required shall be submitted in writing by the manufacturer to the qualifying activity. The manufacturer shall also immediately inspect and/or test power packages currently in production and those awaiting shipment and certify that the deficiency has been corrected prior to further shipment. In addition, the manufacturer shall rework all of the ram air turbine power packages in the production lot represented by the qualification verification test sample provided that these samples are returned to the manufacturer within one year of the date of manufacture. This requirement shall in no way prejudice the Government's position with respect to any other warranty stated or implied under the terms of the contract.

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4.7 Test conditions. Unless otherwise specified, each test required by Table I shall be made under the following conditions.

4.7.1 Mounting. The ram air turbine power package shall be mounted with the AC/DC generator and hydraulic pump in the normal operating configuration. The axis of rotation of the test assembly shall be parallel within one degree with the flow direction of the ram air supply.

4.7.2 Ambient temperature. Unless otherwise specified, the test ambient temperature shall be 25°C ± 15°.

4.7.3 Altitude. Unless otherwise specified, the tests shall be conducted at sea level.

4.7.4 Instrumentation tolerances. Unless otherwise specified, the following tolerances shall apply to all tests:

Temperature	+ 2.8°C (air and fluid temperature
Pressure	∓ 1.5%
Speed	∓ 50 rpm
Generator loads	∓ 5%
Transient frequency	∓ 1/4%
Steady State frequency	∓ 1/2%
Input power	∓ 2%
Fluid flow	∓ 1%
Test voltage	∓ 1/4%
Air velocity	∓ 3 knots
Gravitational units	∓ 10%
All other measurements	± 1%

All instruments and equipment shall be calibrated at intervals that will insure the reported data shall have the accuracy specified above.

4.7.5 Balanced phase loading. The load per phase shall be balanced within one percent of the average of the three phases.

4.7.6 Load power factor. Tests shall be conducted at 0.8 lagging to unity power factor.

4.8 Test methods.

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4.8.1 Examination of product. Each ram air turbine power package, subassembly, and part submitted for test shall be examined as the Government Inspector may deem necessary to determine conformance to this specification with respect to material, workmanship, cleanliness, standard parts, simplification, fabrication, electrical connections, threaded parts, finish, interchangeability, marking, installation instructions, dimensions, weight, color, and any other requirements of paragraph 3.

4.8.2 Emergency power package. The ram air turbine and turbine speed control shall be tested as an integral part of the emergency power package. The tests of Table I shall be performed in any logical order except that the calibration test shall be performed before and after the environmental tests and before and after the endurance test.

4.8.2.1 Rated capacity. The ram air turbine power package shall be operated and stabilized at 116, 126, and 187 knots and meet the output requirements of paragraph 3.10.1.2 for these airspeeds.

4.8.2.2 Start and warm-up. The ram air turbine power package shall demonstrate in conjunction with the system of Figure I five (5) successful and successive starts, with no rework to the power package, per paragraph 3.10.1.4.e at each temperature extreme (5°C and 120°C) except that the pump and reservoir for the 5°C test shall be at -55°C. Sufficient temperature measurements shall be made to insure that the system is at a stabilized temperature at -55°C prior to cold starting tests. Demonstrate five (5) starts each in accordance with paragraph 3.10.1.4.a and with 3.10.1.4.b at a system temperature of -40°C, except that the pump and reservoir temperature shall be at -55°C. No warm-up shall be required for proper operation.

4.8.2.3 Ram air turbine governor. Governing shall be demonstrated at the airspeeds, electrical loads, and hydraulic loads specified in Figure 2 that will maintain an rpm range required by paragraph 3.10.1.1.

4.8.2.4 Turbine overload test. The ram air turbine power package shall be operated at an airspeed of 214 knots, ambient pressure of 29.92 in hg. ± 3% and a temperature of 15°C ± 15°C, and conformance with paragraph 3.10.1.3 demonstrated.

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4.8.2.5 Strength test. The structural integrity of the unit and mounting provisions shall be demonstrated by a suitable static test simulating an equivalent airspeed of 662 knots. A turbine burst test shall be conducted to demonstrate the ability of the ram air turbine without generator and pump to withstand 200 percent rated rpm.

4.8.2.6 Calibration test. Before and after the group of environmental tests of paragraph 4.8.7 and before and after the endurance tests of paragraph 4.8.8, the emergency power package shall be calibrated according to the schedule outlined in the following sub-paragraphs. The ambient temperature, installed airspeed, ambient pressure, hydraulic pump load, generator load, generator output frequency, shall be recorded at five minute intervals. The turbine speed control shall demonstrate conformance with paragraph 3.10.4.1 both during steady-state operation at each of the following test conditions and during changes from one test condition to another.

a. Landing condition:
Ambient temperature - 25° ± 10°C
Ambient pressure - 29.92 in. hg. abs ± 3%
Airspeed - 116 knots
Hydraulic and electric load - 10 minutes at zero load; 10 minutes at the normal load as specified in paragraph 3.10.1.2.a.
Reservoir pressure - 85 psig.

b. Low speed condition:
Ambient temperature - 25° ± 10°C
Ambient pressure - 29.92 in hg. abs ± 3%
Airspeed - 126 knots
Hydraulic and electric load - 10 minutes at zero load; 10 minutes at the load specified in paragraph 3.10.1.2.b
Reservoir pressure - 85 psig.

c. Rated electric power condition:
Ambient temperature - 25° ± 10°C
Ambient pressure - 29.92 in. hg. abs. ± 3%
Airspeed - 187 knots
Hydraulic and electrical load - 10 minutes at zero load; 10 minutes at the normal load as specified in paragraph 3.10.1.2.c
Reservoir pressure - 85 psig.

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d. Hydraulic fluid temperature schedule - During the package tests hydraulic fluid temperature shall be varied to insure performance over the operating range according to the following time schedule:

Temperature	Portion of test time
4°C	15%
90°C	70%
120°C	15%

4.8.3 AC power source. The AC generator and regulator shall be tested along with the DC power source over the range of temperatures specified in Figure 3.

4.8.3.1 Rated capacity. The AC power source shall demonstrate conformance with the requirements of paragraph 3.10.1.2 at the airspeeds specified.

4.8.3.2 Voltage regulation. The AC power source shall demonstrate the ability to deliver AC ouput as specified in paragraph 3.10.1.2 and meet the voltage limit requirement of paragraph 3.10.3.1.2. The test load will be varied from zero to rated load.

4.8.3.2.1 Voltage versus frequency. With the power package delivering 1100 watts AC power, the airspeed shall be varied to demonstrate the underfrequency protection requirements of paragraph 3.10.3.1.2.

4.8.3.3 Voltage modulation. The output voltage modulation shall be determined when the generator is operated carrying zero load to rated load for the speeds specified in paragraph 3.10.1.2. The voltage modulation shall meet the requirements of paragraph 3.10.3.1.3.

4.8.3.4 Phase displacement and voltage unbalance. The effects of single phase and unbalanced three phase loads on the balance of the three phase voltages shall be determined at rated load for the speeds specified in paragraph 3.10.1.2 at the unity power factor. The voltage unbalance shall meet the requirements of paragraph 3.10.3.1.4 when subjected to the load unbalance specified.

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4.8.3.5 Wave form. Conformance to the crest factor and harmonic content requirements specified in paragraph 3.10.3.1.5 shall be demonstrated under the conditions of paragraph3.10.1.2.

4.8.3.6 Frequency limits. Conformance to the requirements of paragraph 3.10.3.1.6 shall be demonstrated.

4.8.3.7 Overload. The overload capacity specified in paragraph 3.10.3.1.1.2 shall be demonstrated according to the conditions specified in paragraph 3.10.1.3.

4.8.4 DC power source. Tests on the DC power source shall be conducted along with the AC power source over the range of temperature specified in Figure 3.

4.8.4.1 Rated capacity. The DC power source shall demonstrate conformance with the requirments of paragraph 3.10.1.2 at the airspeeds specified.

4.8.4.2 Voltage regulation. During voltage regulation tests on the AC generator and voltage regulator, the DC voltage output shall remain within the limits of 24 to 30 volts from zero load to rated load under steady state conditions.

4.8.4.3 Ripple voltage. Ripple voltage shall not exceed the requirements of paragraph 3.10.3.2.3 at loads of 1/4, 1/2, 3/4, and rated.

4.8.4.4 Overload. The overload capacity specified in paragraph 3.10.3.2.1.2 shall be demonstrated according to the conditions specified in paragraph 3.10.1.3.

4.8.4.5 Short-circuit capacity. The ability of the DC power source to deliver short circuit current to the requirements of paragraph 3.10.3.2.1.3 shall be demonstrated. The DC source shall be operated at rated load to stabilized temperature before this test.

4.8.5 Dielectric strength. The ability of the AC/DC power source to withstand a dielectric test according to the requirements of paragraph 3.10.3.3 shall be demonstrated.

4.8.6 Hydraulic power source. The hydraulic pump shall be tested in accordance with paragraph 4.5 of MIL-P-7858. The pump shall be run at rated speed instead of

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3700 rpm. The following exception to paragraph 4.5 of MIL-P-7858 shall apply: air bleed shall be in accordance with paragraph 4.3.2.1.9.6 of MIL-P-19692 with the exception that air ingestion shall be accomplished at least once every 24 hour period during the 200 hours of endurance testing. High temperature tests shall be in accordance with MIL-H-8775. The pump case must be proof pressure tested for 500 psig.

4.8.6.1 Rated capacity. The hydraulic power source shall demonstrate conformance with the requirements of paragraph 3.10.1.2 at the airspeeds specified.

4.8.6.2 Cold run. The pump when cold soaked to -55°C must demonstrate the capability of being started and accelerated to rated rpm within one second with a maximum oil inlet temperature of + 20°C. This shall be demonstrated with a 5/8 inch outside diameter, .049 inch wall thickness aluminum inlet line twenty feet in length. Hydraulic fluid shall be supplied from a reservoir at a maximum pressure of 85 psig static to the pump inlet line. The reservoir and fluid therein will be cold-soaked to -55°C. The pump outlet line will connect to the inlet line approximately twelve inches downstream of the reservoir.

4.8.7 Environmental. The entire ram air turbine power package shall be subjected to the following tests according to the requirements of paragraph 3.11.

4.8.7.1 Temperature shock. The emergency power package to be tested shall be placed within a test chamber wherein, a temperature of 85oC is maintained. The equipment shall be exposed to this temperature for a period of 4 hours, at the conclusion of which, and within 5 minutes, the equipment shall be transferred to a chamber having an internal temperature of -40°C. The equipment shall be subjected to this temperature for a period of 4 hours. This constitutes 1 cycle. The number of complete cycles shall be three. The duration of exposure at each extreme temperature shall not be less than that specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the equipment shall be removed from the test chamber, returned to standard temperature and within a period of 1 hour the performance check of 4.8.7.9 shall be conducted.

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4.8.7.2 Electromagnetic interference. Test shall be conducted in accordance with paragraph 4.5 of MIL-E-81910. The ram air turbine and the hydraulic pump can be excluded.

4.8.7.3 Sand and dust. The dust test shall be in accordance with paragraph 4.2 of MIL-E-81910.

4.8.7.4 Shock. The shock test shall be in accordance with paragraph 4.7 of MIL-E-81910.

4.8.7.5 Vibration. The vibration test shall be in accordance with paragraph 4.9 of MIL-E-81910 except that operation during test is not required.

4.8.7.6 Salt fog. The salt fog test shall be in accordance with paragraph 4.6 of MIL-E-81910.

4.8.7.7 Fungus. The fungus test shall be in accordance with paragraph 4.3 of MIL-E-81910.

4.8.7.8 Humidity. The humidity test shall be in accordance with paragraph 4.4. of MIL-E-81910.

4.8.7.9 Performance check after environmental tests. A performance check will be conducted according to 4.8.2.6 after the following environmental tests:

a. Temperature shock (4.8.7.3)
b. Humidity (4.8.7.8) and salt fog (4.8.7.6)
c. Vibration (4.8.7.5)
d. Sand and dust (4.8.7.1)
e. Shock (4.8.7.4)

4.8.8 Endurance. The service life of the ram air turbine power package shall be 250 hours and shall be demonstrated by a suitable endurance test. The unit shall be operated 100 hours with rated load as specified in paragraph 3.10.1.2. Each of the three conditions will be demonstrated for a minimum of 30 hours. During the 100 hour test, a minimum of 200 deployments into the airstream shall be simulated to demonsrate start reliability. Upon completion of the test, no part of the power package shall exhibit signs of structural weakness or excessive wear.

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4.8.9 Failure definition. The ram air turbine power package is considered failed if during or after any test herein, the unit will not meet specified performance requirements, or if the maintenance of the unit is impeded, or if the unit requires the replacement of components.

5. PREAPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The power package shall be preserved and unit packaged in accordance with Specification MIL-P-116 and MIL-STD-794 as specified by the procuring activity.

5.2 Marking of shipments. Marking shall be as specified in MIL-STD-129. In addition, each container shall be marked with the following information, if specified by the procuring activity:

Machine nomenclature
Federal stock number
Manufacturer's part number
Manufacturer's serial number
Contract number
Date of manufacture
Quantity
Manufacturer or trademark
Contractor (if different from manufacturer)

6. NOTES

6.1 Intended use. The ram air tubine power packages covered by this specification are intended for new and retrofit use as electrical and hydraulic power sources in aircraft. They function, upon failure of the primary power sources, as an emergency source of three phase 115 VAC and 28 VDC electrical power and 7.6 gpm, 3000 psi hydraulic power.

6.2 Drawings. The weights and dimensions as shown on applicable drawings shall include base weight and dimensions for shipping. Weight and dimensions, including any additional equipment necessary to obtain the performance required in the specification shall be included on the drawings.

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6.3 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. Manufacturer's part number
c. Levels of preservation, packaging, packing, and marking (see section 5)
d. Quantity

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products, which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturer's are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360; however information pertaining to qualification of products may be obtained from the Naval Air Test Center, Systems Engineering Test Directorate, Electrical Systems Branch (SY60), Patuxent River, Maryland 20670.

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Navy - AS
Project No. 6115-N367

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TABLE I

TEST METHODS

Test	Paragraph	Acceptance	Qualification Power Package No. 1	Qualification Power Package No. 2	Qualification Verification
Examination of product	4.8.1	X	X	X	X
Rated capacity	4.8.2.1	X	X	X	X
	4.8.3.1	X	X	X	X
	4.8.4.1	X	X	X	X
	4.8.6.1	X	X	X	X
Start and warm up	4.8.2.2		X		X
	4.8.6.2		X		X
OverLoad	4.8.2.4		X	X	X
	4.8.3.7			X	X
	4.8.4.4			X	X
Voltage regulation	4.8.3.2	X	X	X	X
	4.8.4.2	X	X	X	X
Voltage modulation	4.8.3.3			X	X
Phase displacement and voltage unbalance	4.8.3.4		X		X
Wave form	4.8.3.5			X	X
Frequency limits	4.8.3.6	X	X	X	X
Ripple voltage	4.8.4.3		X		X
Short circuit capacity	4.8.4.5			X	X
Dielectric strength	4.8.5	X	X	X	X
Temperature Shock	4.8.7.1			X	
Electromagnetic interference	4.8.7.2			X	
Sand and dust	4.8.7.3			X	
Shock	4.8.7.4			X	
Vibration	4.8.7.5			X	X
Salt fog	4.8.7.6		X		X
Fungus	4.8.7.7		X		X
Humidity	4.8.7.8			X	
Endurance	4.8.8		X		

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TABLE 2

TUBE ASSEMBLY INFORMATION FOR FIGURE 1

ITEM NO.	PART NO.	QTY.	DESCRIPTION OR LENGTH	OD	WALL THICKNESS	NO. BENDS	BEND ANGLES
1 **	215-22125-125	1	5.94	.625	.049	2	1-56° 2-54°
2 **	215-22125-411	1	18.19	.625	.049	3	1-66° 2-136° 3-30°
3 **	215-32100-402	1	42.44	.625	.049	6	1-38° 2-55° 3-14° 4-90° 5-47° 6-33°
4 **	215-32100-244	1	52.88	.625	.049	6	1-27° 2-24° 3-39° 4-40° 5-28° 6-30°
5 **	215-32100-313	1	49.81	.625	.049	9	1-33° 2-48° 3-96° 4-50° 5-51° 6-38° 7-36° 8-26° 9-22°
6 **	215-22125-126	1	6.25	.375	.022	3	1-32° 2-57° 3-43°
7 *	215-22125-613	1	.375	.375	.022	0	-----
8 *	215-22125-614	1	10.56	.500	.028	2	1-61° 2-60°

ALL BEND RADII TO BE 3D

* MATL: CRES 304 PER CVA 1-404
** MATL: 6061T6 PER MIL-T-7081

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TABLE 2 (CONT)

ITEM NO.	PART NO.	QTY.	DESCRIPTION OR LENGTH	OD	WALL THICKNESS	NO. BENDS	BEND ANGLES
9 *	215-22125-546	1	6.06	.500	.028	2	1-32° 2-29°
10 *	215-32100-403	1	35.81	.500	.028	6	1-38° 2-55° 3-14° 4-90° 5-47° 6-26°
11 *	215-32100-245	1	17.00	.500	.028	6	1-45° 2-32° 3-38° 4-40° 5-20° 6-26°
12 *	215-32100-611	1	17.00	.500	.028	1	1-30°
13 *	215-32100-611	1	19.31	.500	.028	3	1-90° 2-45° 3-45°
14 **	215-32100-314	1	20.69	.500	.035	5	1-18° 2-40° 3-92° 4-40° 5-45°
34 **	215-32100-315	1	10.44	.750	.065	1	1-90°

ALL BEND RADII TO BE 3D

* MATL: CRES 304 PER CVA 1-404
** MATL: 6061T6 PER MIL-T-7081

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TABLE 3

COMPONENT INFORMATION FOR FIGURE 1

ITEM NO.	PART NO.	QTY	DESCRIPTION	REMARKS
15	215-22307-2	1	Swivel Joint	Dumont Engr. Co. Long Beach, Calif. Code Ident 97928 P/N 140355 5/8 tubes at ports Press drop of 10 psi at rated flow of 10.5 gpm at 100°F
16	202-21604W10	1	45° Elbow	
17	MS21924D10	2	Union	
18	215-22307-1	1	Swivel Joint	Dumont Engr. Co. Long Beach, Calif. Code Ident 97928 P/N 140350 Press Drop of 35 psi at rated flow of 6.7 gpm at 100°F. 3/8 tubes at ports.
19	MS21916D8-6	1	Reducer	
20	202-21603W8	3	90°Elbow	
21	MS21924D8	1	Union	
22	202-21607W8	2	Elbow	
23	215-32484-1	1	Filters Assembly	Aircraft Porus Media Inc. Glen Gove, New York Code Ident 01414 P/N AC-3258-8HV (Conforms to MS28895-8C) Pressure drop of 15 psi at 90°F (clean filter) at rated flow of 6.0 gpm. By-pass relief valve cracking pressure = 100 ± 10 psi.

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TABLE 3 (CONT)

ITEM NO.	PART NO.	QTY	DESCRIPTION	REMARKS
24	215-22131-3	1	Flow Sens. Press. Regulator	Vickers Inc. Detroit, Michigan Code Ident 62983 P/N EA-50002-24 (See Figure 5 for flow and pressure characteristics)
25	MS21902D8	1	Union	
26	MS21902D6	1	Union	
27	MS21902D10	1	Union	
28	CVC4233-6	1	"O" Ring	
29	215-32367-2	1	Manifold	
30	CVC4233-8	4	"O" Ring	
31	AN6289D8	2	Nut	
32	CVC4230-8	2	B/U Ring	
33	CVC4233-10	1	"O" Ring	
35	202-21603W12	1	Elbow 90°	
36	AN6289D12	1	Nut	
37	CVC4233-12	1	"O" Ring	
38	CVC4230-12	1	B/U Ring	
39	215-32108-2	1	Reservoir	
40	CVC4120-8	1	Check Valve	

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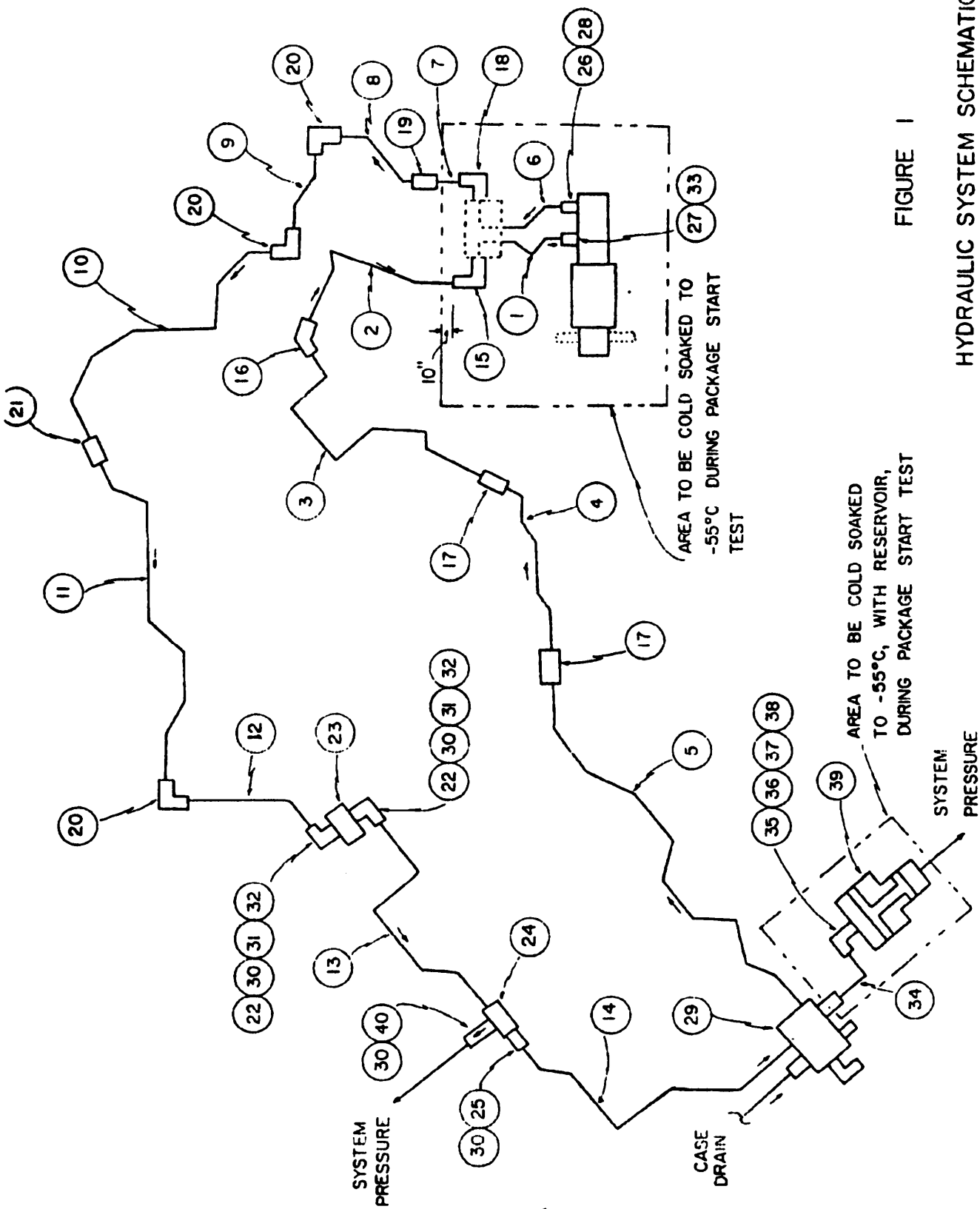


FIGURE 1 HYDRAULIC SYSTEM SCHEMATIC

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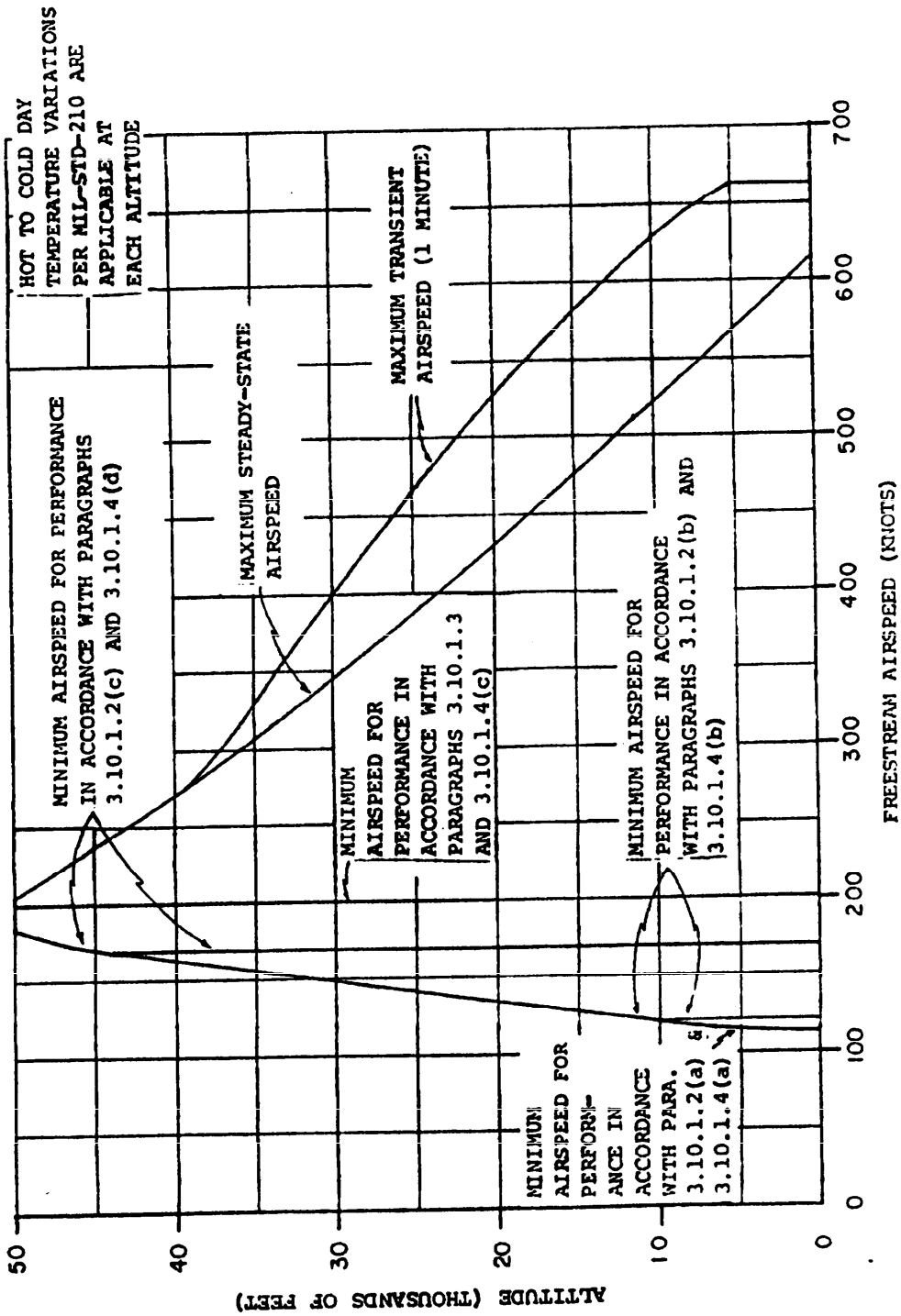


FIGURE 2

POWER PACKAGE OPERATING AIRSPEED ENVELOPE

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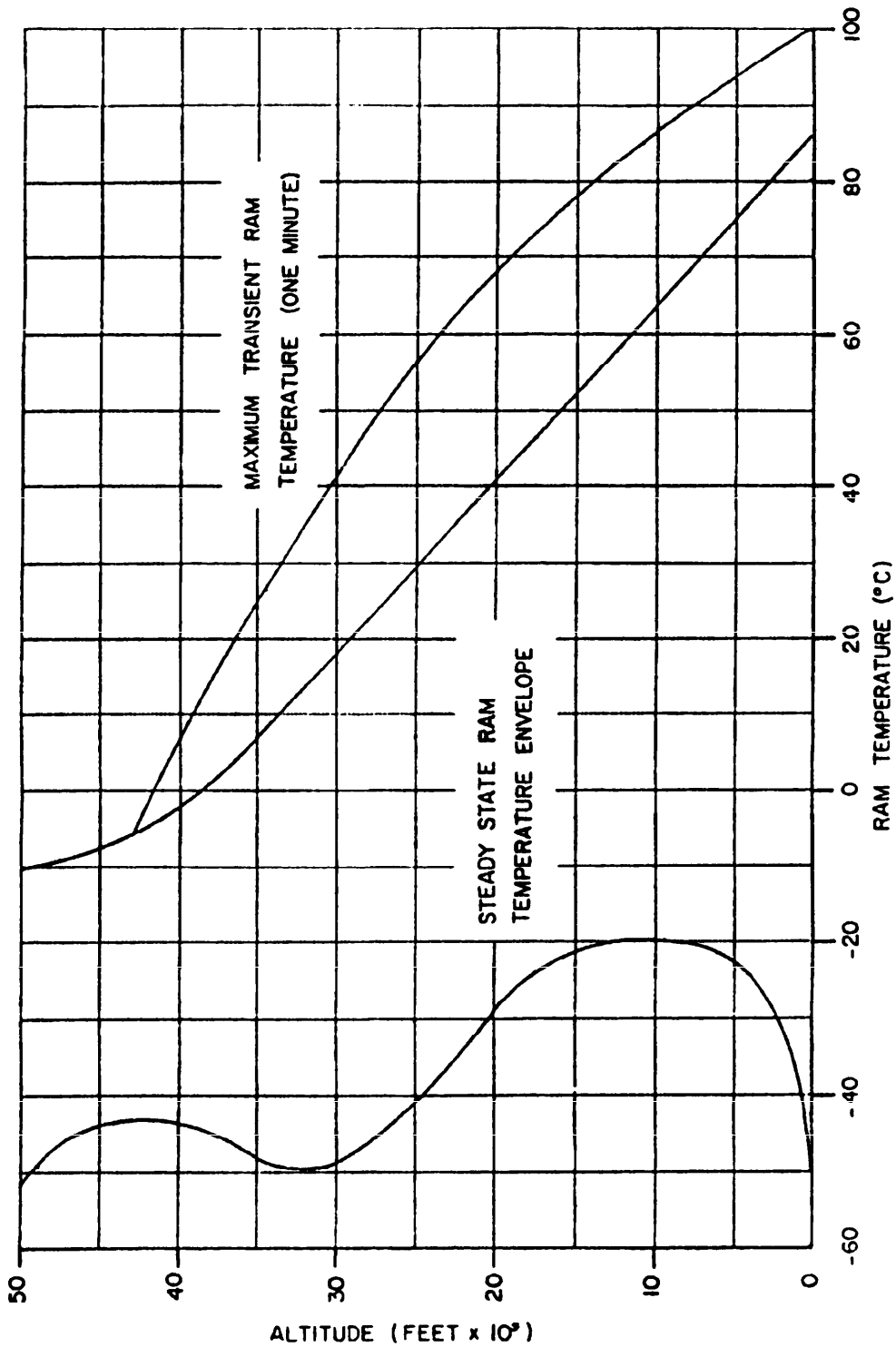


FIGURE 3

POWER PACKAGE RAM AIR TEMPERATURE ENVELOPE

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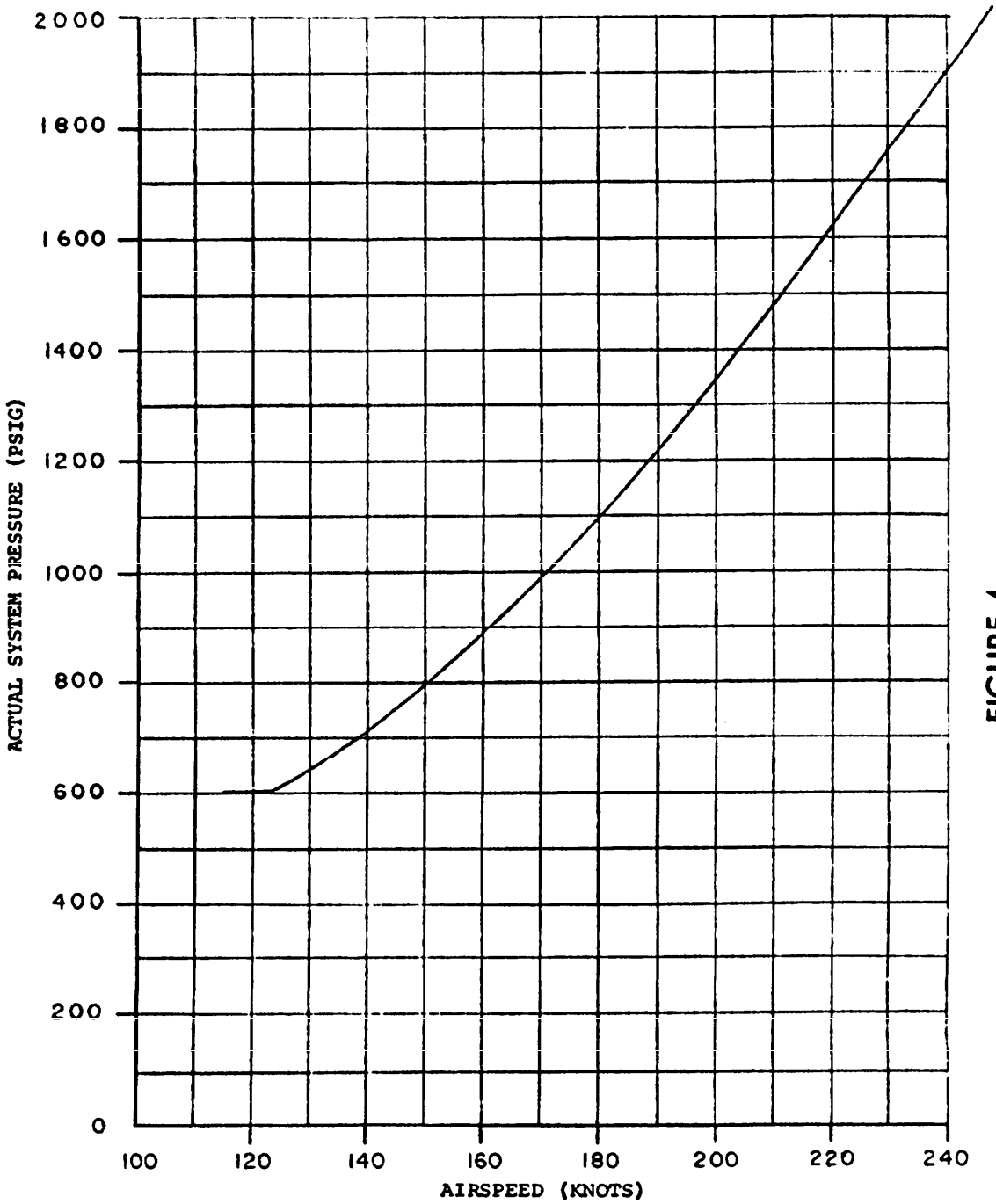


FIGURE 4
REQUIRED FLOW SENSITIVE PRESSURE REGULATOR DISCHARGE PRESSURE VERSUS AIRSPEED

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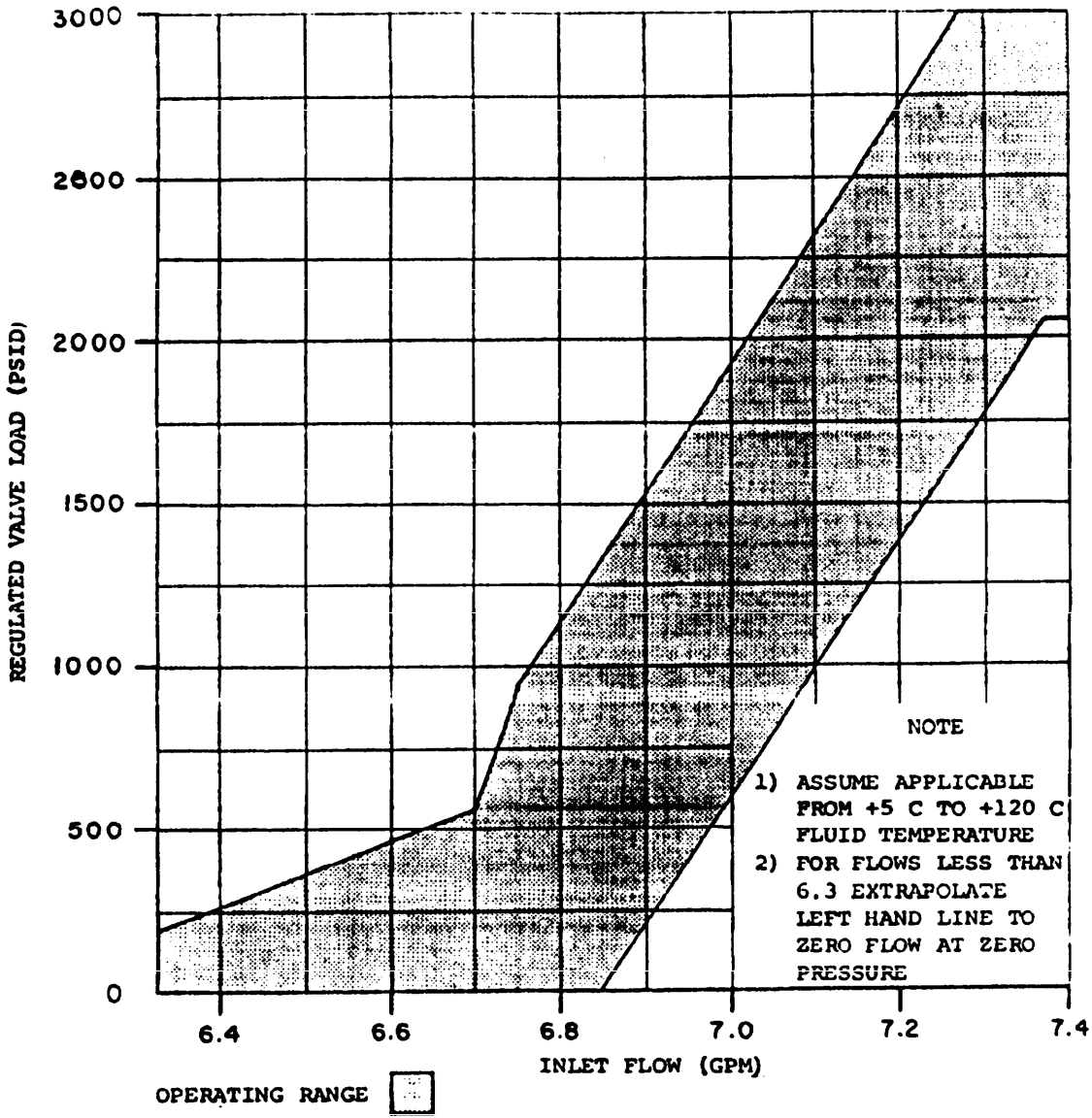


FIGURE 5 FLOW SENSITIVE PRESSURE REGULATOR CHARACTERISTICS

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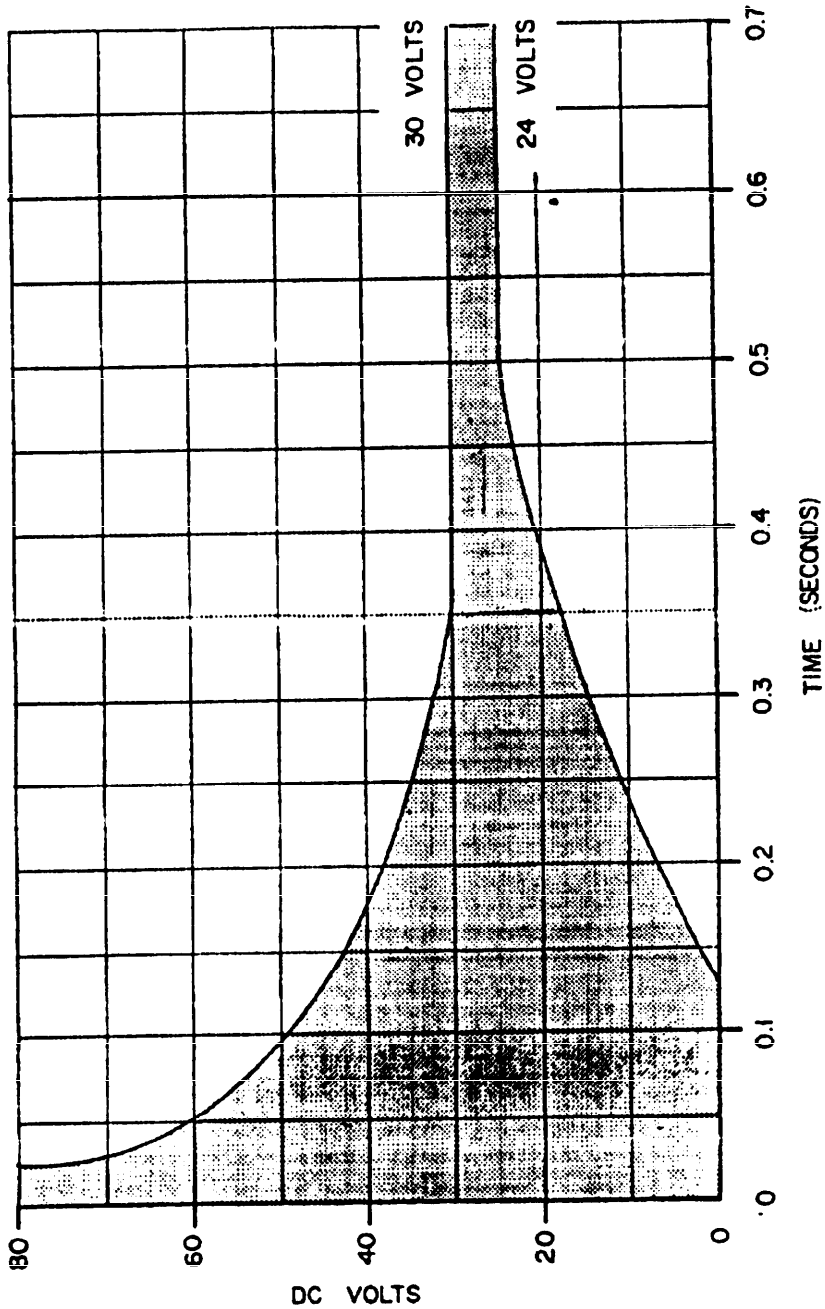


FIGURE 6

TRANSIENT DC VOLTAGE LIMITS

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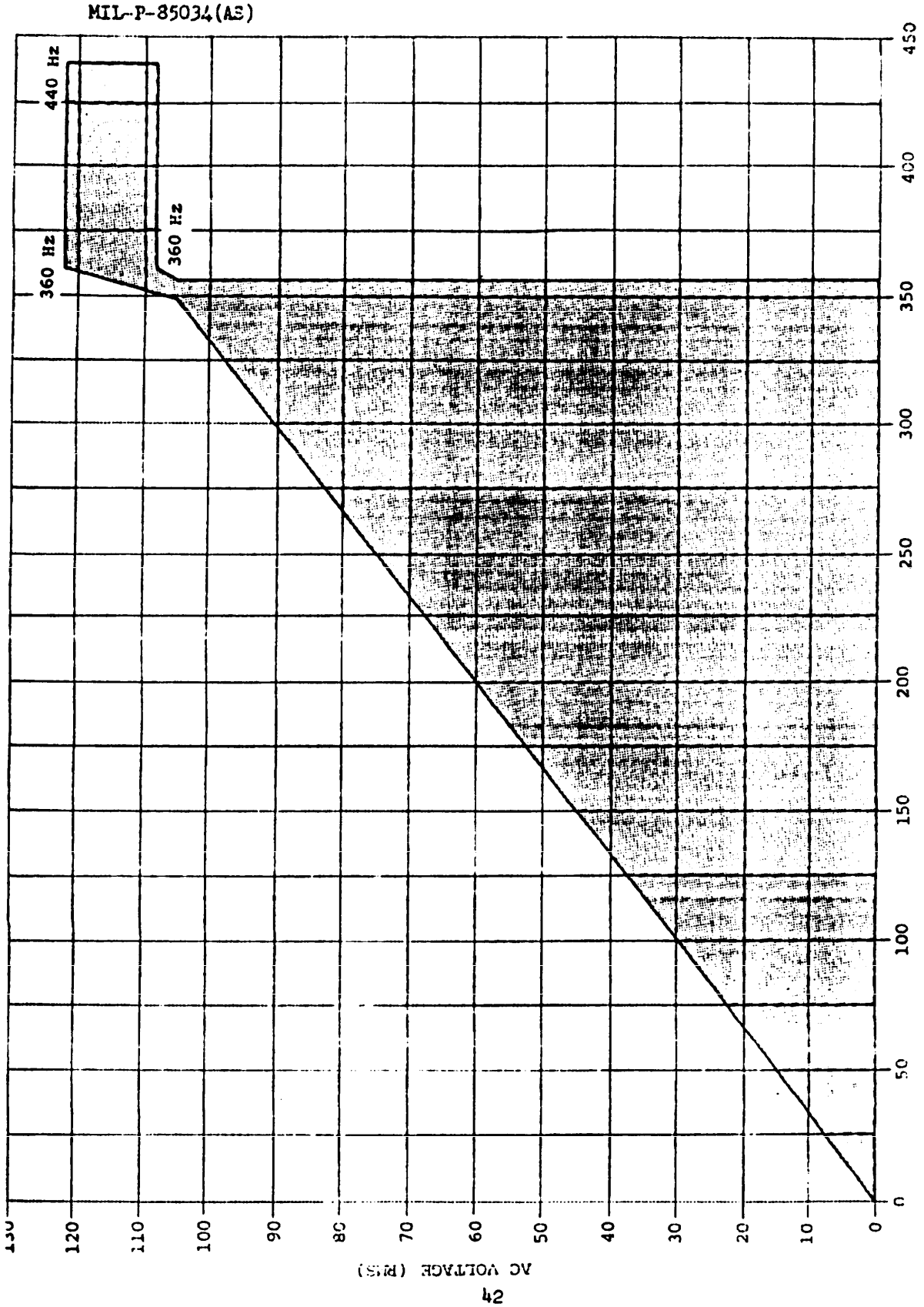


FIGURE 7
FREQUENCY VERSUS VOLTAGE RELATIONSHIP

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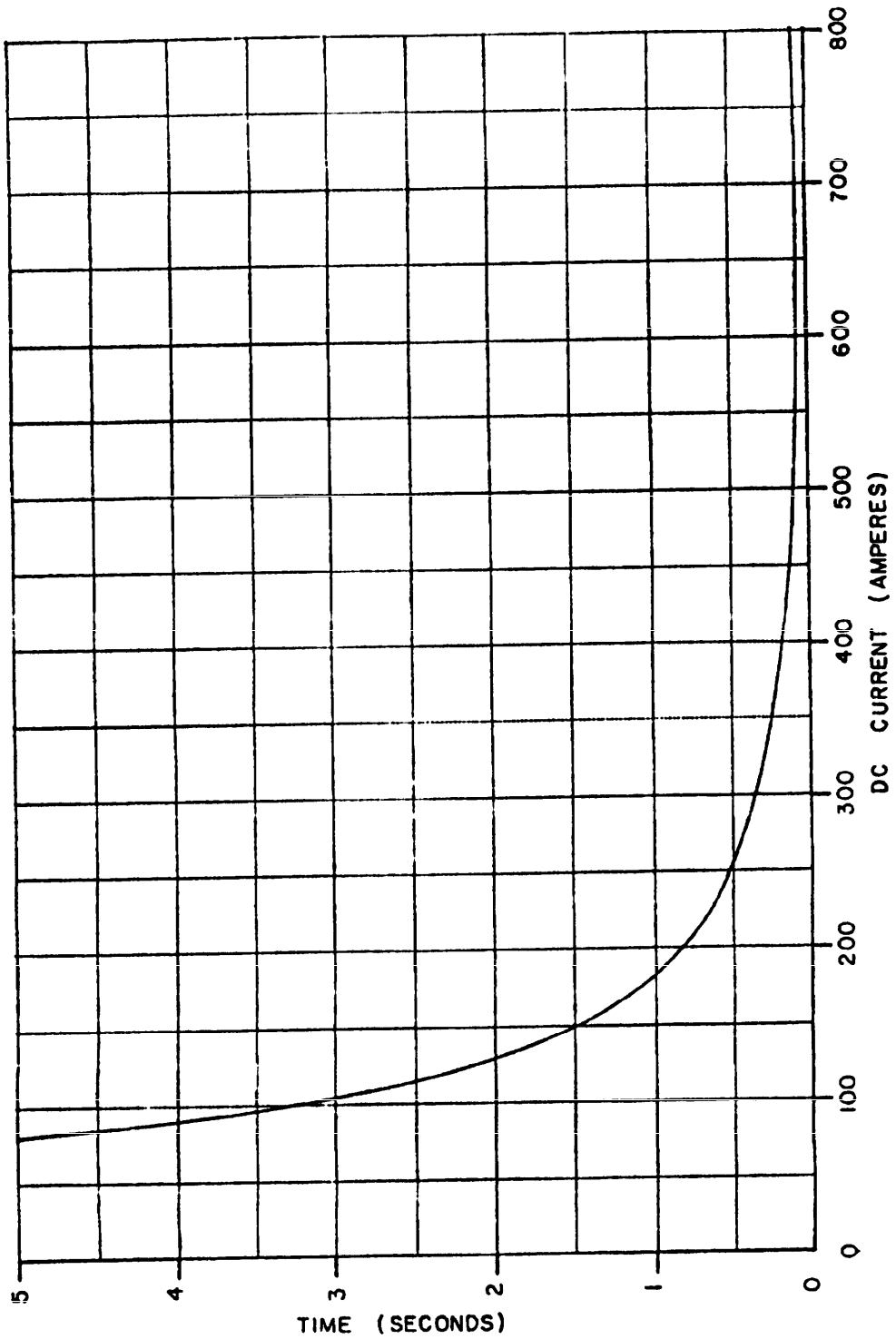


FIGURE 8

SHORT CIRCUIT DC CURRENT RESPONSE

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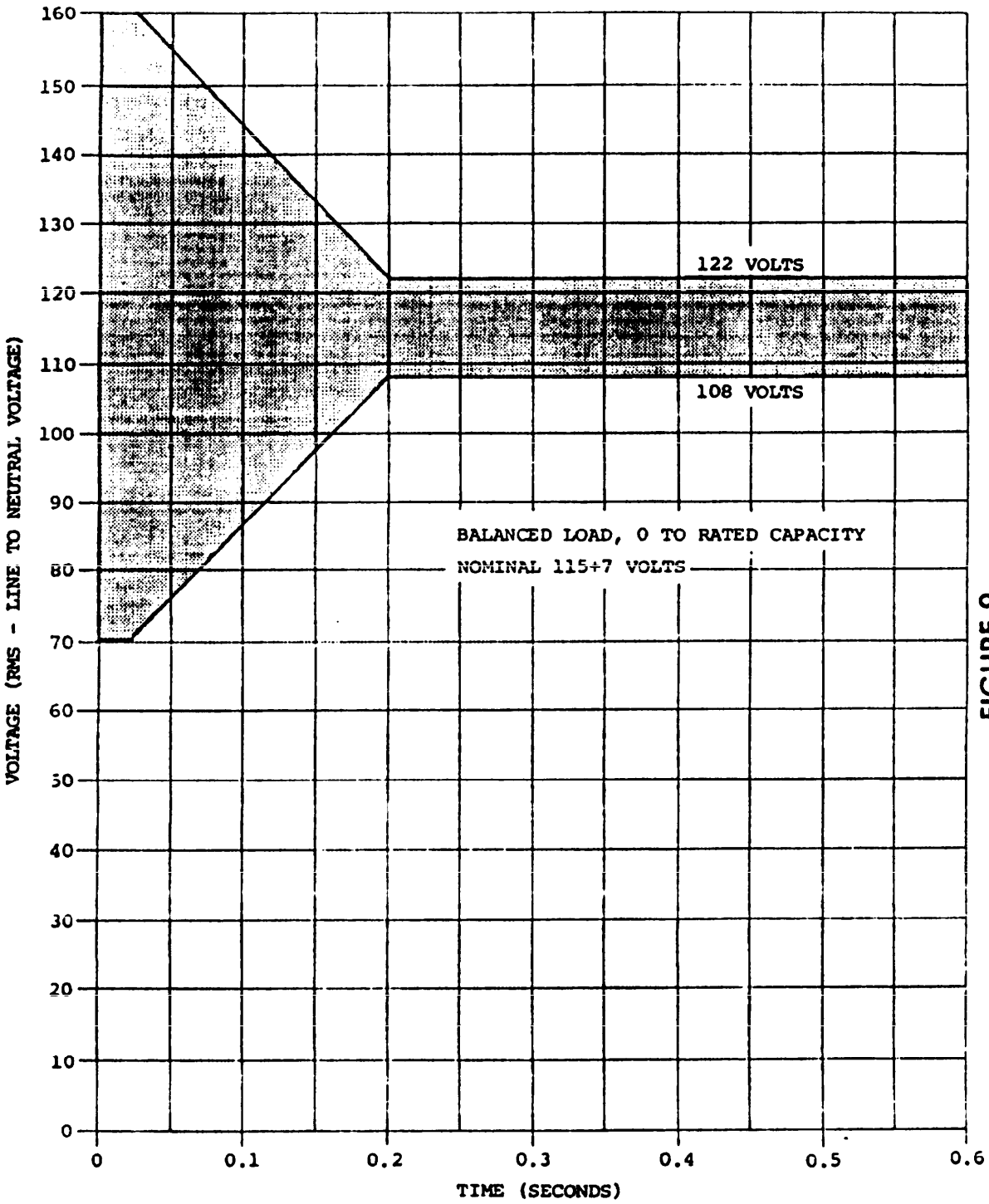


FIGURE 9 TRANSIENT AND STEADY STATE VOLTAGE LIMITS

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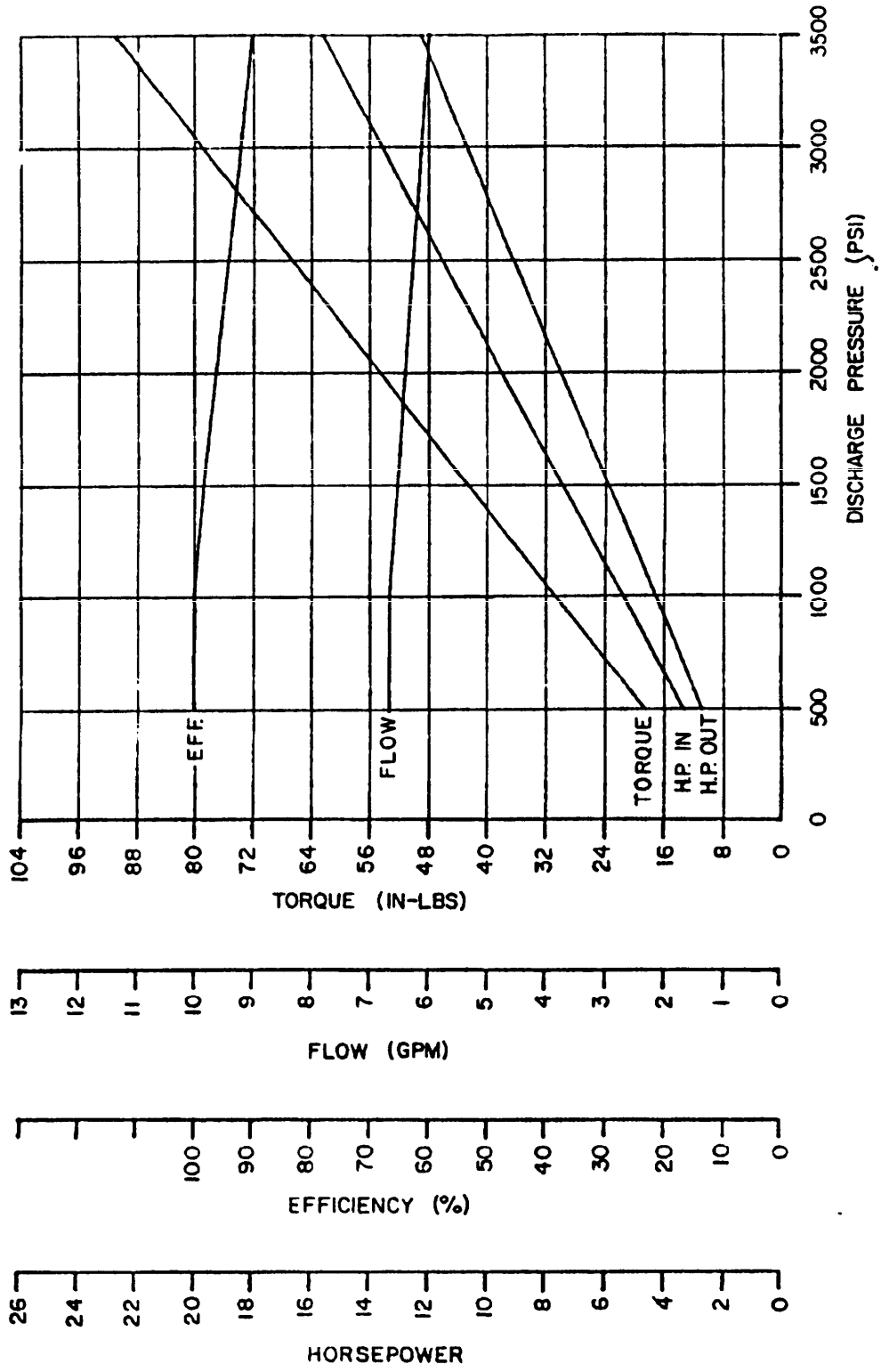


FIGURE 10

HYDRAULIC PUMP PERFORMANCE CHARACTERISTICS

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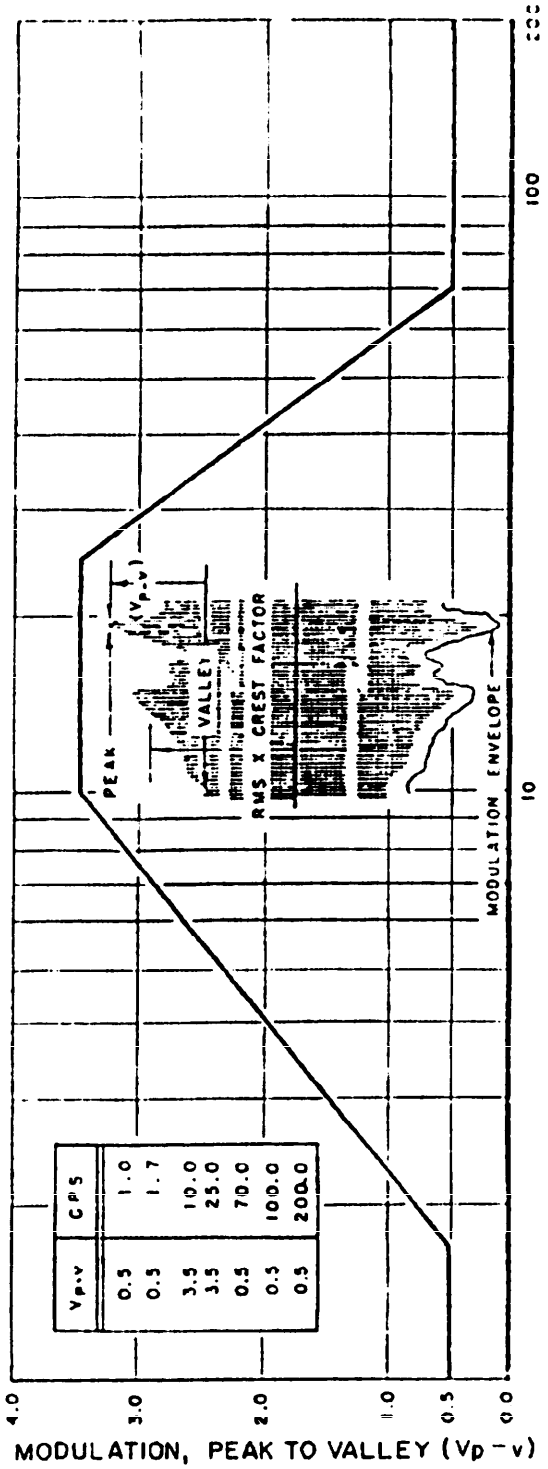


FIGURE 11

FREQUENCY CHARACTERISTICS OF AC VOLTAGE MODULATION ENVELOPE

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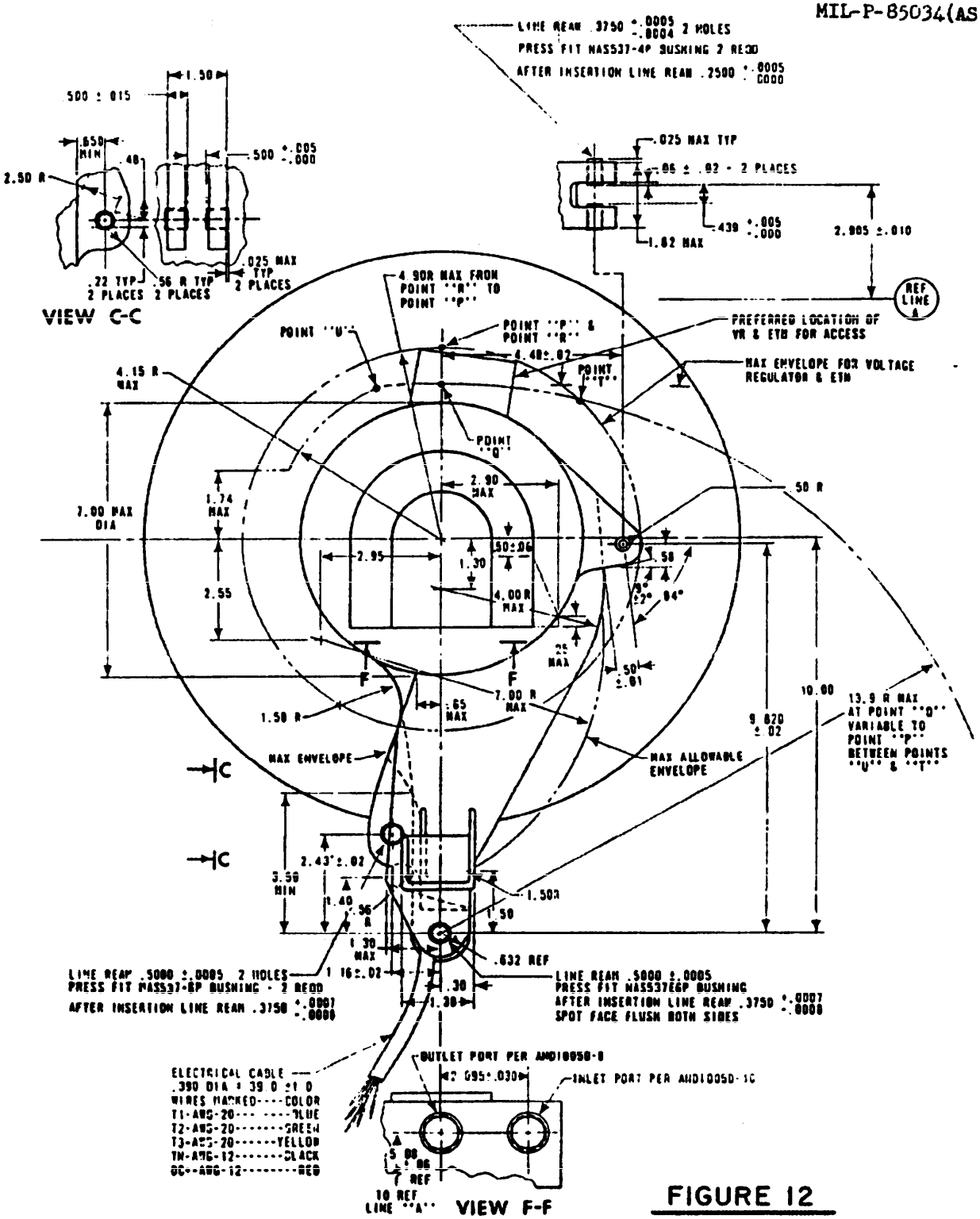


FIGURE 12

EMERGENCY POWER PACKAGE
REAR VIEW ENVELOPE DIMENSIONS

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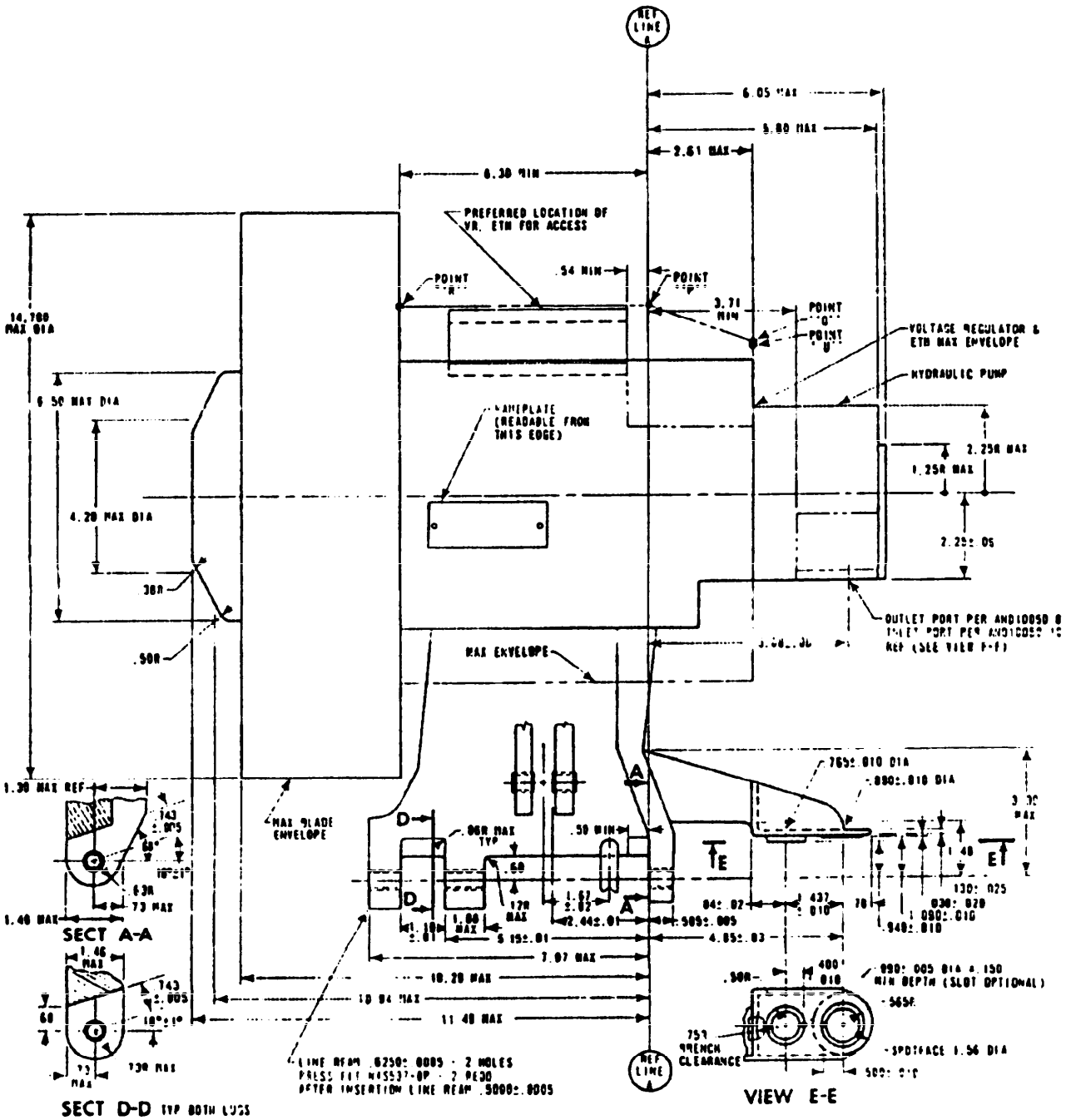


FIGURE 13

EMERGENCY POWER PACKAGE SIDE VIEW ENVELOPE DIMENSIONS

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